

Preprocessing

COMP3009J: Information Retrieval

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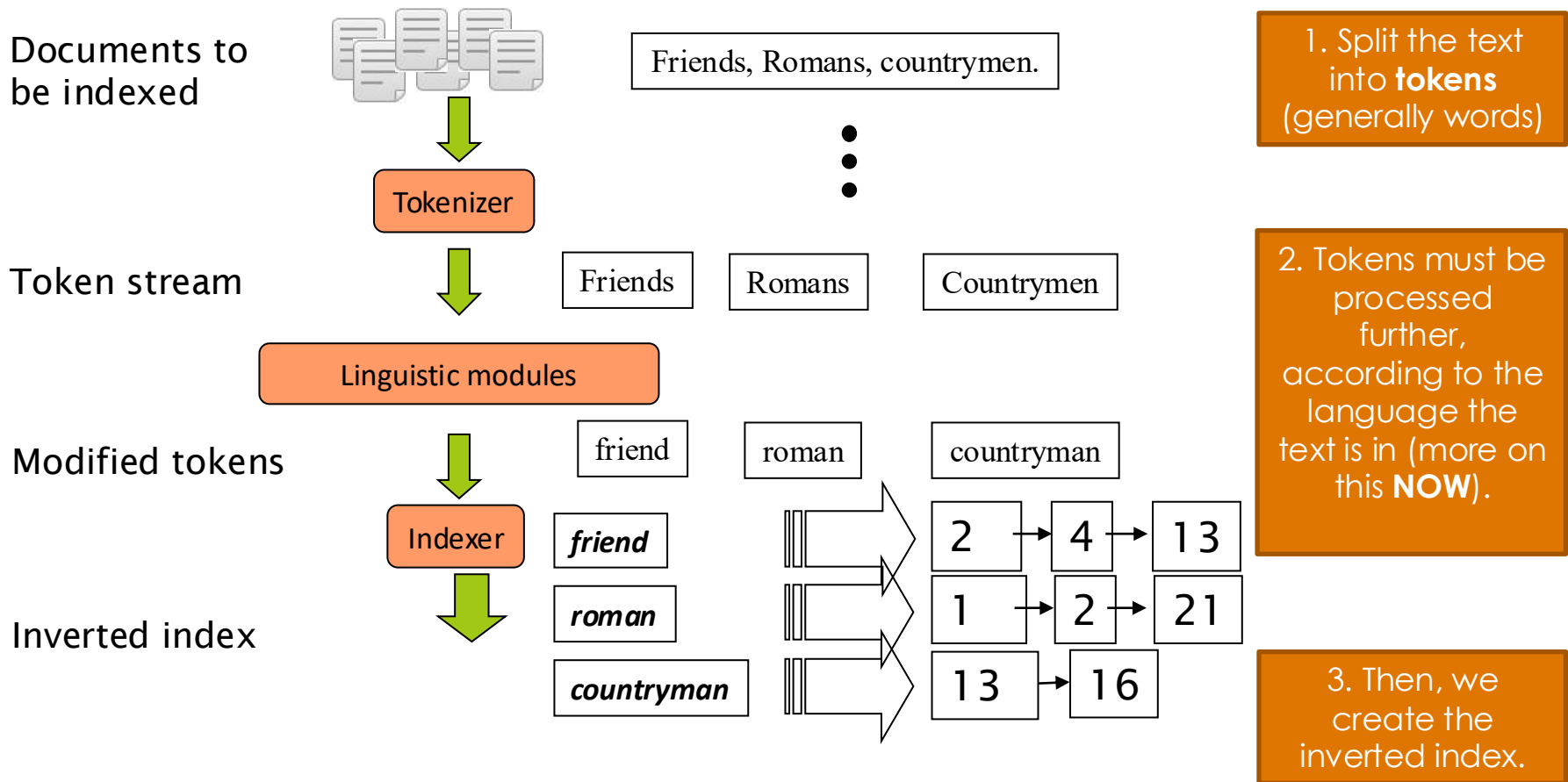
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Preprocessing

- **Aim:**

- Investigate some of the challenges of dealing with natural language when creating index terms.

Remember: creating an Index



Tokenisation (or “Tokenization” if you’re American)

- ▣ **Input:** “Friends, Romans and Countrymen”
- ▣ **Output:** Tokens:
 - ▣ *Friends*
 - ▣ *Romans*
 - ▣ *Countrymen*
- ▣ A **token** is an instance of a sequence of characters. In the previous lecture we said they were similar to words, but they are not the same.
- ▣ Each such token is now a candidate for storing as an index entry, after **further processing**. We refer to this as **preprocessing** as it occurs before queries are processed by the system.
 - ▣ When we store a token in an index, we call it a **term**.
 - ▣ **Note:** a term is not always a real word, as we shall see later.
- ▣ How can we turn tokens into terms?
 - ▣ Libraries are available to deal with most of these situations.

Tokenisation

- Issues in tokenisation:
 - ***Finland's capital*** → *Finland* AND *s*? *Finlands*? *Finland's*?
 - ***Mercedes-Benz*** → ***Mercedes*** and ***Benz*** as two tokens?
 - ***state-of-the-art***: break up hyphenated sequence.
 - ***lowercase, lower-case, lower case*** ?
 - It can be effective to get the user to put in possible hyphens
 - ***San Francisco***: one token or two?
 - How do you decide it is one token?

Numbers

■ 3/20/91

Mar. 12, 1991

20/3/91

■ 55 B.C.

■ B-52

■ *My PGP key is 324a3df234cb23e*

■ (800) 234-2333

- Often have embedded spaces

- Older IR systems may not index numbers

- But often very useful: think about things like looking up error codes/stacktraces on the web

- Will often index “meta-data” separately

- Creation date, format, etc.

Tokenisation: language issues

- French
 - **L'ensemble** → one token or two?
 - **L ? L' ? Le ?**
 - Want **l'ensemble** to match with **un ensemble**
 - Until at least 2003, it didn't on Google
 - Internationalisation!
- German noun compounds are not segmented
 - **Lebensversicherungsgesellschaftsangestellter**
 - 'life insurance company employee'
 - German retrieval systems benefit greatly from a **compound splitter** module
 - Can give a 15% performance boost for German

Tokenisation: language issues

- Chinese and Japanese have no spaces between words:
 - 莎拉波娃现在居住在美国东南部的佛罗里达。
 - Not always guaranteed a unique tokenization.
 - e.g. (烘手)机 vs. 烘(手机)
- Further complicated in Japanese, with multiple alphabets intermingled.



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Katakana Hiragana Kanji Romaji

End-user can express query entirely in hiragana

Tokenisation: language issues

- Arabic (or Hebrew) is basically written right to left, but with certain items like numbers written left to right.
- Words are separated, but letter forms within a word form complex ligatures.

استقلت الجزائر في سنة 1962 بعد 132 عام من الاحتلال الفرنسي.

← → ← → ← start

- ‘Algeria achieved its independence in 1962 after 132 years of French occupation.’
- With Unicode, the way it is visually displayed is complex, but the stored form is straightforward.

Normalisation

- We may need to “normalise” tokens so that they become terms in the same form.
 - e.g. we want *USA* and *U.S.A.* to match.
- Some common approaches:
 - Changing all tokens to lowercase.
 - Even if something should have uppercase letters in it, users often type in lowercase anyway.
 - Delete full stops to form terms: *USA*, *U.S.A.* → *usa*
 - Delete hyphens:
anti-discriminatory, *antidiscriminatory* → *antidiscriminatory*

Thesauri and soundex

- Do we handle synonyms and homonyms?
 - E.g., by hand-constructed equivalence classes
 - **car** = **automobile** **color** = **colour**
 - We can rewrite to form equivalence-class terms
 - When the document contains **automobile**, index it under **car-automobile** (and vice-versa)
 - Or we can expand a query
 - When the query contains **automobile**, look under **car** as well
- What about spelling mistakes?
 - One approach is Soundex, which forms equivalence classes of words based on phonetic heuristics (i.e. it indexes terms using their sounds rather than their spelling).

Conclusions

- When creating an index from a set of tokens, several challenges are presented.
- These challenges depend on the language that the document collection is written in.
- Several approaches have been proposed to address these problems.
 - For many approaches, a tradeoff is required, as they do not improve retrieval in all circumstances.